

1.1 Data, Information, Knowledge, and Wisdom

1. Which of the following is the textbook definition of data?

- (a) Raw facts that cannot give meaning on their own (b) Processed data in a meaningful way (c) The wise use of knowledge (d) An understanding acquired by integrating information

3. Data is recorded and stored using

- (a) only formulas (b) only audio files (c) numbers, texts, and images (d) only graphs

5. Words such as a person's name are only represented by

- (a) text (b) numbers (c) images (d) formulas

7. When we say that "+251..." is a person's phone number, the data becomes

- (a) data (b) wisdom (c) information (d) procedure

9. When daily sales data are aggregated by weeks of the month, the data becomes information because it

- (a) shows whether the sales are increasing or decreasing (b) is raw and unprocessed (c) is stored in a database (d) can be deleted easily

11. Which of the following is a form in which information can be represented?

- (a) numbers only (b) sounds only (c) table, line graph, bar graph, or formulas (d) binary digits only

13. According to the textbook, knowledge is acquired by integrating information from sources such as

- (a) reading books, engaging in different activities, discussing with other people, and observing (b) only listening to the radio (c) only watching films (d) only browsing social media

15. The textbook describes knowledge as more of a

2. Facts such as "100", "250", "age", and "Ethiopia" are considered in the textbook as

- (a) information (b) data (c) wisdom (d) knowledge

4. According to the textbook, which of the following is better represented by numbers than by texts?

- (a) name (b) occupation (c) residence (d) age

6. Which form of data representation has become very popular with the development of satellite technology?

- (a) numbers (b) image (c) texts (d) audio

8. Information is defined in the textbook as

- (a) raw facts that give no meaning (b) a set of instructions (c) a series of steps (d) data processed in a meaningful way to the end users

10. According to the textbook, posing questions such as "who", "what", "when", and "where" helps us to

- (a) delete unnecessary data (b) derive valuable information from the data and make it more useful (c) replace the hardware (d) install new software

12. Knowledge is defined as

- (a) unprocessed raw facts (b) the ability to make decisions without thinking (c) data put into context (d) an understanding acquired by further processing and integrating different sets of information

14. In the textbook's example, when a child applies the information given about riding a bicycle, the information becomes

- (a) data (b) knowledge (c) wisdom (d) raw facts

16. Wisdom is the ability to

- (a) store large amounts of data (b) operate computer hardware (c) design a website (d) make

(a) public library resource (b) hardware device (c) personal possession that you retrieve and apply when you need it (d) software package

17. In the data-information-knowledge-wisdom hierarchy, which of the following is the highest level of human understanding?

(a) wisdom (b) knowledge (c) information (d) data

wise decisions and judgments using one's knowledge

18. Which statement about the DIKW pyramid is correct?

(a) data is at the top and wisdom is at the bottom
(b) data is at the bottom and wisdom is at the top
(c) the value decreases as we move up (d) all levels have equal value

1.2 Information Systems and Their Components

19. An information system (IS) is best defined as

(a) a single computer (b) a collection of web portals
(c) a set of interrelated components working together to collect, process, store, and disseminate information (d) a type of e-learning platform

21. According to the textbook, managers use information to make decisions such as

(a) determining the selling price of a product (b) installing software (c) repairing hardware (d) designing a school building

23. In the school IS example, the summarized assessment data is used to show

(a) the teacher's level of experience (b) the school's electricity bill (c) a student's overall performance (d) the number of visitors

25. Hardware includes the digital devices required to perform core information system functions such as

(a) data storage, data processing, and communication (b) hiring and promotion (c) marketing and sales (d) budgeting and auditing

27. Software is best described as

(a) the physical devices of a computer (b) the people who operate the system (c) a set of instructions that commands the computer hardware to do something (d) raw unprocessed facts

29. Where must the various types of data collected by organizations be stored for later use?

(a) in a computer database (b) in the telecommunication network (c) in the

20. Information systems generate different types of information or reports that are used by

(a) only hardware devices (b) only competitors (c) only customers (d) employees in the organization to accomplish different organizational tasks

22. In the school information system example, the system captures student data about different types of

(a) building maintenance (b) assessments such as quizzes, assignments and final examinations (c) teacher salaries (d) food supplies

24. The five major components of an information system are

(a) hardware, software, money, people, and marketing (b) hardware, software, networks, databases, and websites (c) input, output, storage, processing, and memory (d) hardware, software, data/information, people, and procedure

26. Communication among different parts of an information system is facilitated through telecommunication networks that include

(a) only postal mail (b) wired and wireless networks (c) only copper wires (d) only radio broadcasts

28. Which of the following is NOT software, according to the textbook's list of software required for an information system?

(a) operating system (b) anti-virus software (c) database management system (d) the central processing unit

30. According to the textbook, which of the following best describes data or information?

(a) an example of hardware (b) a critical resource or asset that must be properly managed (c) a form of network (d) a software license

organization's procedures (d) on the e-government portal

31. The component of an information system that is a series of steps followed to achieve a desired outcome or goal is

(a) hardware (b) people (c) procedure (d) software

32. Clear procedures in an organization clarify

(a) what data should be captured by the information system (b) who should have access to the data (c) who should have the privilege to modify data (d) all of the above

1.3 Applications (1): E-Government, E-banking, E-learning

33. E-services are defined as

(a) the provision of public services using technology as a mode of service delivery (b) the selling of hardware devices (c) the storage of raw facts (d) the maintenance of networks

35. In the context of e-Government, a portal is

(a) a database of employee salaries (b) a hardware device for data storage (c) a web-based information system that collects information from different sources into a single user interface (d) a series of procedures

37. The e-Government interaction that connects a government with individual citizens is known as

(a) G2C (b) G2B (c) G2G (d) G2E

39. Exchanging information between different government agencies and departments is an example of

(a) G2C (b) G2E (c) G2G (d) B2C

41. In the banking sector, IT is mainly associated with services like

(a) Internet banking, Automated Teller Machine (ATM), and mobile banking (b) checkbooks and passbooks (c) money orders and postal transfers (d) coins and banknotes

43. Which of the following is a challenge of e-banking mentioned in the textbook?

(a) lower cost of operation (b) easy account balance checking (c) digital theft (d) immediate messages about transactions

45. If e-learning is provided over the Internet, it is referred to as

34. E-Government is the use of information systems to provide electronic services to citizens with the purpose of increasing

(a) paper-based procedures (b) efficiency, transparency, and citizen participation (c) manual service delivery (d) the number of government buildings

36. Which of the following is NOT a benefit of e-Government mentioned in the textbook?

(a) establishing trust between governments and citizens (b) reducing cost of running government services (c) facilitating sharing of information between government agencies (d) requiring citizens to visit government offices in person

38. The type of e-Government interaction that deals with the relationship between a government and businesses is

(a) G2C (b) G2B (c) C2C (d) B2B

40. Electronic banking is a form of technology-based banking service that avoids

(a) the use of computers (b) the use of mobile phones (c) electronic signals (d) the exchange of cash, checks, or other types of paper documents

42. Which of the following is a benefit of e-banking mentioned in the textbook?

(a) customers must wait in line at the bank (b) customers can withdraw cash any time using, for example, ATM service (c) high operation costs per unit of service (d) customers must visit the bank for every payment

44. E-learning is a form of learning delivered through

(a) face-to-face classes only (b) postal mail (c) radio broadcasts only (d) use of digital devices

46. According to the textbook, the main components of e-learning include

(a) online learning (b) offline learning (c) correspondence learning (d) broadcast learning

47. Which of the following is a benefit of e-learning?

(a) physical isolation of students from teachers (b) fixed class schedules (c) learners can access and learn at any time or any day (d) high travel and accommodation costs

(a) only teachers and classrooms (b) the Internet, computers or mobile phones, digital learning resources, content developers, and learners (c) only textbooks and pens (d) only school buildings and buses

48. Which of the following is a main disadvantage of e-learning mentioned in the textbook?

(a) flexibility in learning time (b) access to rich multimedia resources (c) reduced educational costs (d) physical isolation of students from teachers and other students

1.3 Applications (2): E-commerce, Telemedicine, Teleconference

49. E-commerce is best defined as

(a) the buying and selling of goods and services over the Internet (b) the use of e-learning (c) the provision of health care online (d) the transfer of government services

51. Customers can easily browse the price of different merchants and choose the least price. This is a benefit of e-commerce known as

(a) avoiding intermediaries (b) fixed pricing (c) increased bargaining power for the customer (d) global competition

53. Which e-commerce advantage means that the Internet allows merchants to expand their customers outside their local area?

(a) global reach (b) avoiding intermediaries (c) convenience (d) lower Internet access

55. Telemedicine was originally introduced to

(a) sell medicines over the Internet (b) replace all city hospitals (c) treat patients who were in remote places, far away from local health facilities, or in areas with shortages of medical professionals (d) reduce the number of doctors

57. Which of the following is an advantage of telemedicine for patients?

(a) overcoming geographical barriers to access health care services (b) increasing travel and hotel expenses (c) requiring patients to visit clinics (d) limiting access to specialized physicians

50. According to the textbook, the actors involved in e-commerce transactions include

(a) only merchants and banks (b) merchants, buyers or customers, banks, and transporters (c) only buyers and sellers (d) teachers and students

52. The e-commerce advantage that allows customers to buy products directly from manufacturers at a cheaper price, avoiding wholesalers and retailers, is called

(a) convenience (b) global reach (c) increased competition (d) avoiding intermediaries

54. Telemedicine refers to

(a) the sale of medicines through vending machines (b) the practice of caring for patients remotely without the caregiver and the patient being in the same physical location (c) medical check-ups done only in hospitals (d) health insurance services

56. Home-use medical devices used with telemedicine can be used to measure a patient's

(a) IQ and memory (b) income and savings (c) age and height only (d) body temperature, glucose levels, and blood pressure

58. Which of the following best describes a teleconference?

(a) a meeting of people who are in the same room (b) a meeting of two or more people separated by distance using electronic communication (c) communication carried out using telegrams (d) communication carried out by sending letters

59. The two popular forms of teleconferencing are

(a) radio and television conferencing (b) email and fax conferencing (c) audio conferencing and video conferencing (d) telephone and telegraph conferencing

60. Video conferencing, unlike audio conferencing

(a) is voice-only communication (b) uses only telephone lines (c) does not allow participants to see each other (d) combines both voice communications and video images